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9 **(a)** By reference to energy transfers, distinguish between electromotive force (e.m.f.) and potential difference (p.d.).

e.m.f. .....

.....

p.d. .....

..... [2]

(b) A circuit is set up as shown in Fig. 9.1.

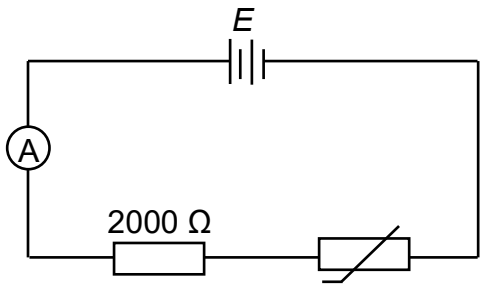

Fig. 9.1

The battery source of emf E is found to provide 2.4×10^5 J of electrical energy to the 2000 Ω resistor and thermistor when a charge of 2.2×10^4 C passes through the ammeter. At room temperature, the thermistor has a resistance of 1800 Ω.

(i) Sketch on Fig. 9.2 the variation with temperature θ of resistance R in a thermistor. [1]

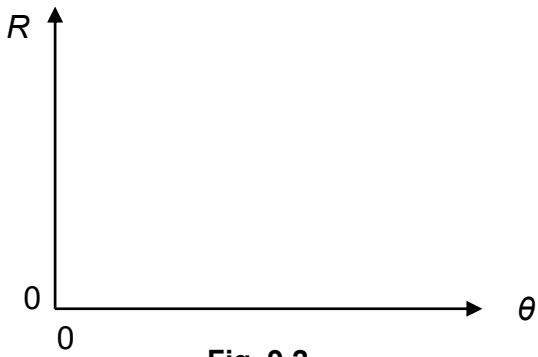

Fig. 9.2

(ii) For the thermistor at room temperature,

1. show that E is 11 V. [1]

2. determine the time taken for a charge of 2.2×10^4 C to pass through the ammeter.

time = ……. s [2]

3. show that the fraction of power dissipated in the thermistor is 0.47. [1]

(c) A uniform resistance wire PQ of length 1.2 m is subsequently connected across the resistor and thermistor, as shown in Fig. 9.3. A sensitive voltmeter is connected between point Y and a moveable contact M on the wire.

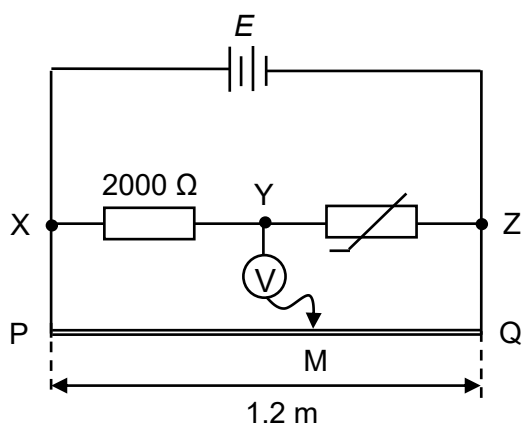

Fig. 9.3

(i) At room temperature, the contact M is moved along PQ until the voltmeter shows zero reading.

Calculate the length of wire between M and Q.

length of wire = ……. m [2]

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(ii) State and explain the effect, if any, on the length of the wire between M and Q for the voltmeter to remain at zero deflection if each of the following changes takes place independently.

1. The thermistor is warmed slightly.

……...

……...

……...

…….. [2]

2. A uniform wire with a bigger cross sectional area is used to replace PQ.

……

……

……

…….... [2]

(d) The circuit shown in Fig. 9.4 is used to compare potential differences.

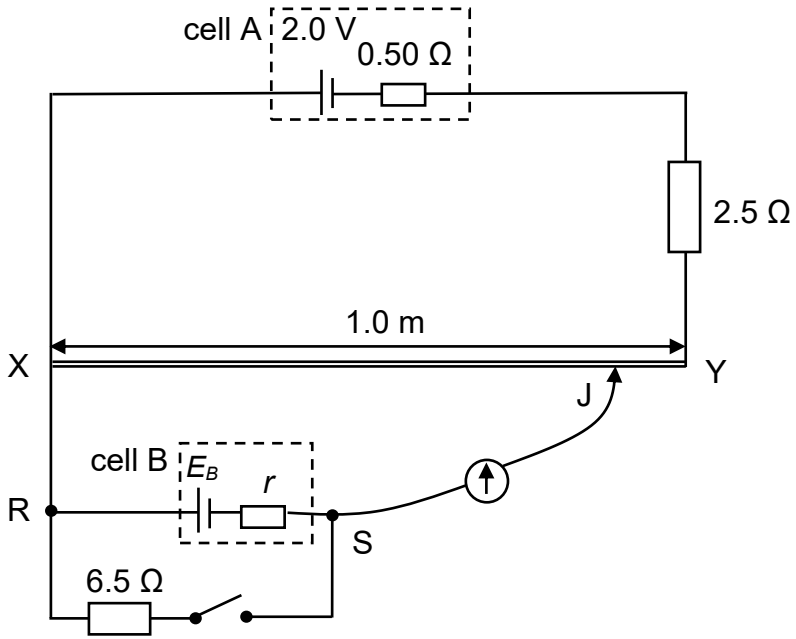

Fig. 9.4

The uniform resistance wire XY has length 1.0 m and resistance 8.0 Ω. Cell A has e.m.f. 2.0 V and internal resistance 0.50 Ω. Cell B has e.m.f. E_B and internal resistance r.

(i) The switch is opened. The galvanometer shows no deflection when the moveable contact J is adjusted so that the length XJ is 0.90 m.

Show that the e.m.f. E_B of cell B is 1.3 V. [2]

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(ii) The switch is now closed.

1. For the galvanometer to show no deflection, contact J has to be adjusted so that length XJ is 0.75 m.

Determine the internal resistance *r* of cell B.

r = …….. Ω [3]

2. A resistor is connected in parallel with the 6.5 Ω resistor.

Deduce how the balanced length XJ would be affected.

……..……..

……..……..

……..……..

…….. [2]